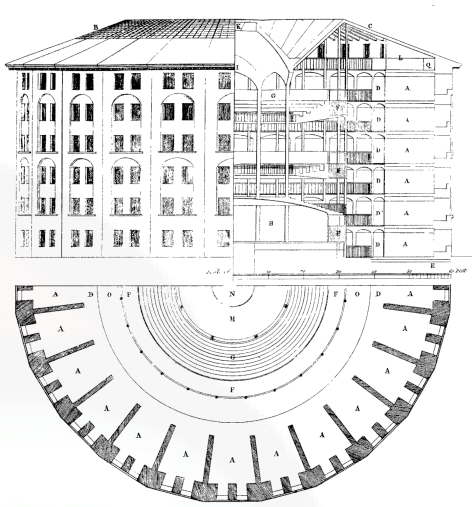
Aiding the Surveillance State:



The University of Virginia Panopticon Information Surveillance System (UVA-PISS) tracks every request in and out of the UVA network, like any good mass surveillance team does. They have recently noticed a large increase in traffic and have hired YOU, the resident Bureaucratic Analyst and Brainiac Algorithmist ( ) to analyze their network data for them.

The UVA-PISS network monitor records server requests. Each request is represented by an integer ID of the device that sent the request. UVA-PISS has already identified through their existing systems network IDs that appear to have abnormal traffic, but they don't know the exact amount of traffic these IDs send over specific intervals. They also suspect that malicious users might be masking their IDs, and so they additionally want information about how many unique IDs make over a suspicious ( ) threshold in a given interval.

In summary, there are two types of queries that are interesting to UVA-PISS.

1. Given a range [L,R] and a specific ID, return the *frequency* of that ID over the range, or the amount of times that ID shows up in the range
2. Given a range [L,R], return the number of *unique* IDs that appear at least *K* times over the range.

Input

The first line contains an integer $N \leq 2^{20}$ denoting the number of times the network was accessed.The second line contains N integers (each between 0 and $(2^{32}-1)$ inclusive), where each integer represents the ID of the computer accessing the server.

The next line contains $Q \leq 2^{16}$ for the number of queries and $1 \leq K \leq N$ for the minimum frequency threshold.

The next Q lines will be either

1. F (for frequency), followed by the ID number of the computer of interest and the left and right indices of our range. I.e., if we want to get the frequency of computer with ID 10 from indices 1 to 3, the query would be F 10 1 3
2. U (for uniqueness), followed by the left and right indices of our range. For example, if we want to get the number of unique computers who have accessed the server at least K times from indices 1 to 3, the query would be U 1 3.

Output

Print Q lines, each containing the answer for a query. The ith line should contain the answer to the ith query.

Sample Input

```
8
1 2 1 3 2 3 1 2
4 2
F 2 0 3
U 0 7
F 1 1 7
U 4 7
```

Sample Output

```
1
3
2
1
```